Major physics theories

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1 Mathematical physics

1.1 Topological structures for physics

1.1.1 Submanifolds

A. Prerequisites: diffeomorphisms and the inverse function theorem

Before studying submanifolds, we need two tools from multivariable calculus that will be used repeatedly. These results concern open subsets of $\mathbb{R}^m$ and require no notion of manifold.

Let us first start with some definitions and reminders on differentiability. All of this is admitted, but proofs can be found in math textbooks on calculus.

Definition: Differentiability.

Let $U \subset \mathbb{R}^m$ be open and $f : U \to \mathbb{R}^k$ a map. We say f is **differentiable at** $p \in U$ if there exists a linear map $L : \mathbb{R}^m \to \mathbb{R}^k$ such that:

$$\lim_{\|h\| \to 0} \frac{\|f(p+h) - f(p) - Lh\|}{\|h\|} = 0 \tag{1}$$

The linear map L is called the **total derivative** of f at p, and is denoted df_p or $Df(p)$. It is the best linear approximation to f near p.

Remark. This is the correct notion of differentiability for maps between spaces of dimension greater than 1. It is strictly stronger than the existence of partial derivatives: a function can have all partial derivatives at a point without being differentiable here. Differentiability at p implies continuity at p and implies the existence of all directional derivatives, but not conversely.

Definition: The Jacobian matrix. If $f : U \subset \mathbb{R}^m \to \mathbb{R}^k$ is differentiable at p, its total derivative df_p is a linear map $\mathbb{R}^m \to \mathbb{R}^k$. In the standard bases of $\mathbb{R}^m$ and $\mathbb{R}^k$, this linear map is represented by the $k \times m$ matrix of partial derivatives:

$$J(p) = \left(\frac{\partial f^i}{\partial x^j}(p)\right)_{\substack{i=1,\dots,k \\ j=1,\dots,m}} = \begin{pmatrix} \frac{\partial f^1}{\partial x^1} & \cdots & \frac{\partial f^1}{\partial x^m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f^k}{\partial x^1} & \cdots & \frac{\partial f^k}{\partial x^m} \end{pmatrix}_p \tag{2}$$

This is called the **Jacobian matrix** of f at p. For a vector $v \in \mathbb{R}^m$, the action of the total derivative is:

$$df_p(v) = J(p)\, v \tag{3}$$

Remark: chain rule. If $g : V \to \mathbb{R}^\ell$ is differentiable at $f(p)$, the composition $g \circ f$ is differentiable at p and:

$$d(g \circ f)_p = dg_{f(p)} \circ df_p \tag{4}$$

In terms of Jacobian matrices: $J_{g\circ f}(p) = J_g(f(p))\, J_f(p)$. This is the coordinate expression of the chain rule.

Definition: Smoothness. A map $f : U \subset \mathbb{R}^m \to \mathbb{R}^k$ is called **smooth**, written $f \in \mathcal{C}^\infty(U)$, if all partial derivatives of all orders exist and are continuous on U. In physics, one often only requires as many derivatives as the calculation needs. We will follow the mathematical convention here for precision.

Definition: Diffeomorphism.

Let $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^k$ be open sets. A map $\Phi : U \to V$ is a diffeomorphism if:

- Φ is a bijection,
- Φ is smooth,
- $\Phi^{-1} : V \to U$ is also smooth.

A diffeomorphism is the natural notion of equivalence in the smooth category: two open sets related by a diffeomorphism are indistinguishable from the point of view of differential calculus.

Remark. A bijective smooth map need not be a diffeomorphism. The map $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$ is smooth and bijective, but $f^{-1}(y) = y^{1/3}$ is not differentiable at 0. Smoothness of the inverse must be checked separately.

Theorem: Inverse Function Theorem. Let $\Phi : U_p \subset \mathbb{R}^m \to \mathbb{R}^m$ be smooth and let $p \in U_p$. If $\det J_\Phi(p) \neq 0$, then there exist open sets U'_p and V'_p such that $p \in U'_p$, $\Phi(p) \in V'_p$ and $\Phi|_{U'_p} : U'_p \to V'_p$ is a diffeomorphism.

In other words: if the best linear approximation to Φ at p (given by the total derivative $d\Phi_p$) is invertible (non-zero determinant of the Jacobian), then Φ itself is locally invertible with smooth inverse.

Proof strategy. The proof proceeds by constructing a contraction mapping $T_y(x) = y + x - \Phi(x)$ whose unique fixed point (guaranteed by the Banach fixed point theorem, using the smallness of $J_\Phi - I_m$ near p) defines $\Phi^{-1}(y)$. Smoothness of the inverse follows from the chain rule applied to $\Phi^{-1} \circ \Phi = \text{id}$. We refer to [1, 3] for the complete proof.

Remark: two other important theorem are corollary of the Inverse Function Theorem:

- **Implicit Function Theorem**: $F : \mathbb{R}^{n+r} \to \mathbb{R}^r$, rank $J(p) = r$. Then dF_p is **surjective**, and the level set $F^{-1}(c)$ is locally a graph over $\mathbb{R}^n$.
- **Immersion Theorem**: $f : \mathbb{R}^n \to \mathbb{R}^{n+r}$, rank $J(p) = n$. Then df_p is **injective**, and locally f looks like the standard inclusion $\mathbb{R}^n \hookrightarrow \mathbb{R}^{n+r}$.

B. The graph view of submanifolds

Differential geometry studies spaces that may be globally curved or complicated, but which locally resemble ordinary Euclidean space. The central idea is revolving around the concept of manifold: **a set that locally looks like** $\mathbb{R}^n$. As its definition involves some extent of formalism, following the step of ref. [2], we'll start by motivating its usefulness in physics by admitting that the Euclidian space $\mathbb{R}^n$ which is a manifold, and we first define submanifolds of that space.

We start from ordinary multivariable calculus and asks: What properties must subsets of $\mathbb{R}^N$ satisfy to be able to define differential calculus only on it? To answer this question, we will define graphs of smooth functions and use it to get a first understanding of what a submanifold actually is.

Suppose we have a subset M^n of $\mathbb{R}^{n+r}$, where n and r are integers :

$$M^n \subset \mathbb{R}^{n+r} \tag{5}$$

We will call $\mathbb{R}^{n+r}$ the ambient space. M^n is an embedded submanifold of $\mathbb{R}^{n+r}$ if locally near any point $p \in M^n$, one can find r functions $f^a : \mathbb{R}^n \to \mathbb{R}$ such that :

$$y^a = f^a_p(x^1, \ldots, x^n) \tag{6}$$

Where $(x^1, \ldots, x^n)$ are n free real coordinates, and $(y^1, \ldots, y^r)$ are dependent real coordinates determined by the r smooth functions $(f^1_p, \ldots, f^r_p)$.

Definition: graph description of a manifold

More precisely, a submanifold is locally defined around a point p of neighborhood $U_p \subset \mathbb{R}^{n+r}$ if :

$$\exists f_p = (f_p^1 \dots f_p^r) : \mathbb{R}^n \to \mathbb{R}^r / M \cap U_p = \{(x, f_p(x))/x \in \pi(U_p)\} \tag{7}$$

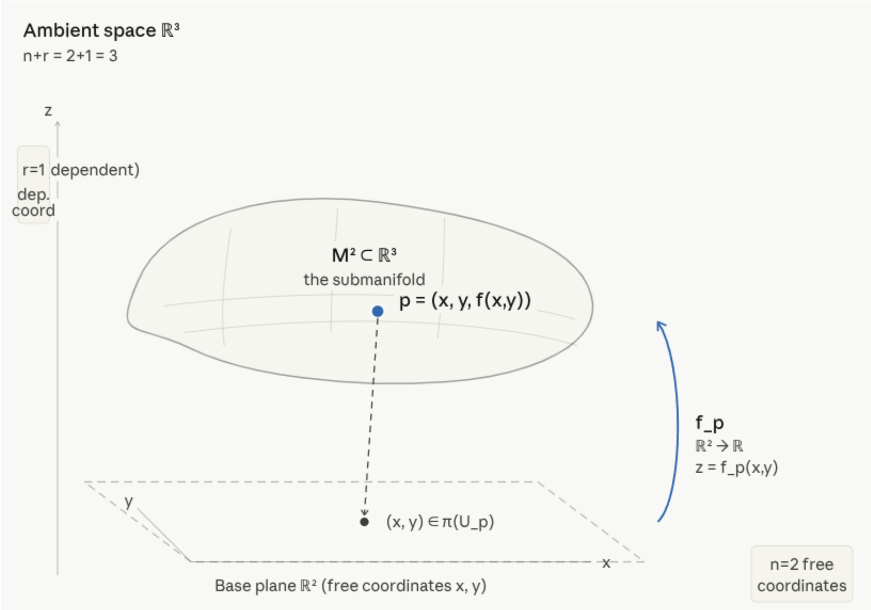

Figure 1: Sketch of a submanifold of $\mathbb{R}^3$: its dimension is 2 (2 free coordinates (x,y)) while the codimension is 1. π is just the projection of a point near p to the n free coordinates, here to the (x, y) plane. Picture generated by A.I., feel free to suggest others if they seem more adapted.

Where $\pi : R^{n+r} \to R^n$ projects onto the n free coordinates. The r smooth maps f_p^a provides a graph representation of the submanifold in the vicinity of one of its point p. n is called the dimension of the submanifold (as these are the free remaining, unconstrained coordinates) while r, the number of constraints or dependent coordinates is called the codimension in the ambient space. The important geometric point to understand is that given $n + r$ independent coordinates, the r maps f_p smoothly constrain r coordinates so that $(x^1, \dots, x^n, y^1, \dots, y^r)$ becomes an $n-$dimensional family of vectors and remain in the neighborhood of p.

Example: A Curve in $\mathbb{R}^2$. Consider $y = f(x)$. This defines a one-dimensional object inside two-dimensional space. The ambient dimension is 2, but the manifold dimension is 1, because only one variable is free. Once x is chosen, y is fixed. Thus, there is only one degree of freedom. f provides a graph representation of the curve, which turns out to characterize it globally.

Example: A Surface in $\mathbb{R}^3$. Now consider $z = f(x, y)$. This defines a surface in $\mathbb{R}^3$. The ambient dimension is 3, but the manifold dimension is 2, because two variables are free, and the third is constrained. Again, the manifold dimension counts the number of independent local parameters.

A manifold does not have to be globally a graph. The word locally matters. The classic example is the circle: $x^2 + y^2 = 1$. Globally, this cannot be written as: $y = f(x)$, because each value of x corresponds to two possible values of y. However, locally this works perfectly: whether we define it for the upper semicircle $y = +\sqrt{1 - x^2}$ or for the lower semicircle $y = -\sqrt{1 - x^2}$, we can find **local maps** constraining variables on the circle. Note that different local graph representations may be required around different points of the same manifold.

We have so far given sufficient conditions to define the subset M^n as a manifold. Now that we have a rough understanding of what a submanifold is, let's try to characterize it better. To this end, we need to introduce level sets and tangent spaces.

C. Level set description of submanifolds

Definition: constraints
We start by introducing a smooth map over a neighborhood of $p \in \mathbb{R}^{n+r}$ denoted U_p:

$$F_p : U_p \subset \mathbb{R}^{n+r} \to \mathbb{R}^r. \tag{8}$$

We call F_p the vector of constraints. Each coordinates of F_p are a map $F_p^a : \mathbb{R}^{n+r} \to \mathbb{R}$ with $a \in \{1, \dots, r\}$ and defines a constraint.
Definition: level set
We introduce :

$$S = F_p^{-1}(c) = \{x \in \mathbb{R}^{n+r} : F_p(x) = c\} \tag{9}$$

Where c is a vector of $\mathbb{R}^r$. Each of its coordinate is an imposed value for one constraint:

$$x \in F_p^{-1}(c) \Leftrightarrow \begin{pmatrix} F_p^1(x) \\ \dots \\ F_p^r(x) \end{pmatrix} = \begin{pmatrix} c^1 \\ \dots \\ c^r \end{pmatrix} \tag{10}$$

S is called a level set (or locus in ref. [2]): it is defined as the set of points such that each constraint (coordinates of F_p) satisfy some value (coordinates of c).
Example: The Sphere. Define $F(x, y, z) = x^2+y^2+z^2$. Then the sphere is $F^{-1}(R^2) = \{(x, y, z) : x^2 + y^2 + z^2 = R^2\}$. Thus, the sphere appears naturally as a constraint surface. As F has only one coordinate, there is only one constraint to this submanifold: it is of dimension 2, codimension 1.

The ambient space $\mathbb{R}^{n+r}$ possesses $n + r$ degrees of freedom. If one imposes r independent constraints $F^a(x) = c^a$, each independent equation removes one degree of freedom. Thus, the remaining number of independent parameters becomes $(n+r)-r = n$. But what does "independent" constraints mean ?
To answer this independency question, we must study the variations of the vector of $\mathbb{R}^r$ they form: F_p. First, let's consider any curve on U_p, the set of points in the neighborhood of p:

$$\begin{aligned} x : \mathbb{R} &\to U_p \\ t &\mapsto x_p(t) \end{aligned} \tag{11}$$

That are such that $x(0) = p$. The variations of the constraints F_p along this curve is its total derivative:

$$\frac{d}{dt} F_p(x(t))|_{t=0} = dF_p(\dot{x}(0)) = J(p)\dot{x}(0) \tag{12}$$

The partial derivatives of F_p^a are the coordinates of the Jacobian matrix:

$$J^{ij} = \left(\frac{\partial F_p^i}{\partial x^j}\right) = \begin{pmatrix} \frac{\partial F^1}{\partial x^1} & \dots & \frac{\partial F^1}{\partial x^{n+r}} \\ \dots & \dots & \dots \\ \frac{\partial F^r}{\partial x^1} & \dots & \frac{\partial F^r}{\partial x^{n+r}} \end{pmatrix} \tag{13}$$

When we defined the total derivative df_p of a map $f : U \subset \mathbb{R}^m \to \mathbb{R}^k$, we described it as a linear map $\mathbb{R}^m \to \mathbb{R}^k$. This is correct but conflates two distinct roles of $\mathbb{R}^m$: as a space of **points** (where p lives), and as a space of **velocity vectors** at p (where $\dot{x}(0)$ lives). For open subsets of $\mathbb{R}^m$ this distinction is invisible: both are just $\mathbb{R}^m$. But it becomes essential on a submanifold, so we introduce it now.
Definition: tangent space of $\mathbb{R}^m$ at p. Let $p \in \mathbb{R}^m$. The **tangent space** $T_p\mathbb{R}^m$ is the set of all velocity vectors of smooth curves through p:

$$T_p\mathbb{R}^m = \{\dot{x}(0)/x : \mathbb{R} \to \mathbb{R}^m \text{ smooth},\ x(0) = p\} \tag{14}$$

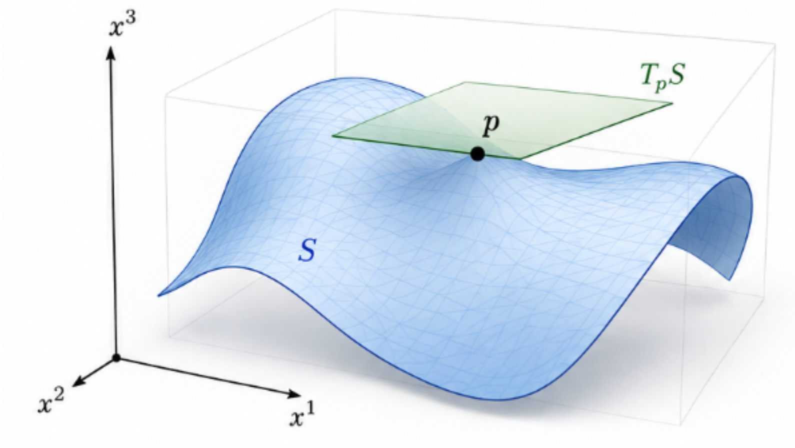

Figure 2: Illustration of the concept of tangent space. Here, the gray box denotes the ambient space, $\mathbb{R}^3$. The blue curve is a submanifold. At $p \in S$, the tangent space sketched as the green plane is a vector space made of all the tangent vector at p. The tangent vectors are obtained by considering the total derivative of trajectories on S crossing p at p. Picture generated by AI, feel free to suggest better illustration for the concept of submanifolds and tangent space.

Since $x(t) = p + tv$ is a valid curve for any $v \in \mathbb{R}^m$, every vector is a tangent vector and $T_p\mathbb{R}^m \cong \mathbb{R}^m$. The tangent space is thus just a copy of $\mathbb{R}^m$ **attached at the point** p, carrying the interpretation of directions rather than positions.

With this language, the total derivative dg_p for a function $g : \mathbb{R}^m \to \mathbb{R}^h$ is more precisely a linear map:

$$dg_p : T_p\mathbb{R}^m \to T_{g(p)}\mathbb{R}^k \tag{15}$$

It takes a velocity vector at p in the domain and returns the corresponding velocity vector at $f(p)$ in the target (the velocity of the image curve $f(x(t))$ at $t = 0$). In coordinates, this is still just $J(p)\,v$, but the interpretation is now geometric: df_p maps directions at p to directions at $f(p)$.

The reason for this language becomes clear on a submanifold $S \subset \mathbb{R}^{n+r}$: a point $p \in S$ still has an ambient tangent space $T_p\mathbb{R}^{n+r} \cong \mathbb{R}^{n+r}$, but not all directions at p stay on S. The tangent space $T_pS \subsetneq T_p\mathbb{R}^{n+r}$ captures only those velocity vectors $\dot{x}(0)$ of curves x that remain on S. It is a strict subspace of dimension $n < n + r$, and will soon show how regular value theorem is precisely what allows identifying it.

Proposition: The tangent space near p of the level set is included in the kernel of the Jacobian matrix:

$$T_pS \subset \ker J(p) \tag{16}$$

Proof

Let us consider a trajectory x defined in eq. 11 near p and belonging to the level set S. Each of its images $x(t) \in U_p \subset \mathbb{R}^{n+r}$ must satisfy $F_p(x(t)) = c$. Let us differentiate :

$$\frac{d}{dt}F_p(x(t)) = \frac{d}{dt}c = 0 \tag{17}$$

Using the matricial representation we introduced with the Jacobian, this equation becomes

$$J\dot{x} = 0 \tag{18}$$

Therefore, any curve in the level set S such that $x(0) = p$ admits a **tangent curve** near p are such that $dF_p(\dot{x}(t)) = 0$, and thus belong in the **kernel of the Jacobian (equivalently, of the linear map $\mathbf{dF_p}$)** at p.

■

We can now clearly answer what independency of constraints means: when does imposing r constraints $F^a(x) = c^a$ actually remove exactly r degrees of freedom? The answer lies in the rank of the Jacobian $J(p)$.

Recall that $J(p)$ is an $r \times (n+r)$ matrix representing the linear map $dF_p : T_p\mathbb{R}^{n+r} \to T_{F(p)}\mathbb{R}^r$. Its rank is exactly what measures how many of the r constraints are genuinely independent near p.

Definition: Regular point and regular value. A point $p \in S = F^{-1}(c)$ in the level set is called a regular point of F if:

$$\operatorname{rank} J(p) = r \tag{19}$$

That is, if the linear map at p, dF_p, is surjective. The value c is called a regular value of F if every point in the level set $S = F^{-1}(c)$ is regular.

The geometric meaning is immediate: if $\operatorname{rank} J(p) = r$, then the r rows of $J(p)$ are linearly independent. But these rows are exactly $\nabla F^a = (\partial_{x_1} F^a, \ldots, \partial_{x_{n+r}} F^a), a \in \{1, \ldots, r\}$ meaning they are the derivative of the r constraint functions F^a. Qualitatively, from this results on their variation ∇F^a, the constraints F^a are genuinely pulling in r independent directions in the ambient space at p. Each independent constraint kills exactly one degree of freedom. A more formal way of proving this is invoking the rank-nullity theorem applied to dF_p:

$$\dim\ker J(p) = \dim \mathbb{R}^{n+r} - \operatorname{rank} J(p) = (n+r) - r = n \tag{20}$$

So the kernel has exactly dimension n at a regular point. Since we showed $T_pS \subset \ker J(p)$, **the tangent space at any regular point has at most dimension** n. The inclusion is true for every level set, even singular ones. Equality requires the regularity condition.

Theorem: Implicit function theorem

We already introduced this theorem in the reminders, as a consequence of the inverse function theorem. Here, we state it more rigorously to prove the equivalence between level set and graph views of submanifold.

Consider a point $p \in \mathbb{R}^{n+r}$ of neighborhood U_p and let $F : U_p \subset \mathbb{R}^{n+r} \to \mathbb{R}^r$ be smooth, $F(p) = c$, and suppose $\operatorname{rank} J(p) = r$. Then, there exist open set $V \subset \mathbb{R}^n$ and $W \subset \mathbb{R}^r$ with $p \in V \times W \subset U$ and a unique smooth map $f : V \to W$ such that

$$F(x, y) = c \Leftrightarrow y = f(x) \quad \forall x, y \in V \times W \tag{21}$$

In other words, near p, the level set S is exactly the graph of f.

Proof

As $\mathbb{R}^{n+r} \cong \mathbb{R}^n \times \mathbb{R}^r$, we can write $u \in \mathbb{R}^{n+r}$ as $u = (x, y)$ with $x \in \mathbb{R}^n, y \in \mathbb{R}^r$. Introducing $p = (x_0, y_0) \in \mathbb{R}^{n+r}$ of neighborhood U_p, we define the map:

$$\begin{aligned} \Phi : U_p &\to \mathbb{R}^{n+r} \\ (x, y) &\mapsto (x, F(x, y) - c) \end{aligned} \tag{22}$$

Since $\operatorname{rank} J(p) = r$, the $r \times (n+r)$ Jacobian of F $J(p)$ has r independent columns. The coordinates can be reordered in a way such that the $r \times r$ bottom right $\frac{\partial F}{\partial y}$ block is invertible. Using such reordering, the Jacobian matrix at p of Φ thus is:

$$J_\Phi(p) = \begin{pmatrix} I_n & 0 \\ \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \end{pmatrix} \tag{23}$$

This is a block lower triangular matrix, with $det J_\Phi(p) = det\frac{\partial F}{\partial y} \neq 0$. By the inverse function theorem, Φ is a local diffeomorphism near p: there exists a neighborhood U_p of p on which Φ has a smooth inverse Φ^{-1}. In particular if we consider $(x, y) \in \mathbb{R}^{n+r}$ such that $F(x, y) = c$, then $\Phi(x, y) = (x, 0)$ thus $(x, y) = \Phi^{-1}(x, 0)$ and, therefore there exist a smooth function $f : \mathbb{R}^n \to \mathbb{R}^r$ such that $y = f(x)$. The smoothness of f follows from the fact that $f(x)$ is the second component of $\Phi^{-1}(x, 0)$: since Φ^{-1} is smooth by the inverse function theorem, and projection onto a component is smooth, f is smooth.

■

The rank condition $\text{rank}\, J(p) = r$ guarantees that the constraints $F^a = c^a$ are non-degenerate at p: the Jacobian pushes the r constrained directions in genuinely independent directions. The implicit function theorem then says: you can always solve locally for r coordinates in terms of the remaining n; turning the level set view into the graph view. The two descriptions are equivalent near any regular point.

Theorem: Regular Value Theorem.

Let $F : U \subset \mathbb{R}^{n+r} \to \mathbb{R}^r$ be smooth and c a regular value of F. Then $S = F^{-1}(c)$ **is a smooth submanifold of dimension** n, with tangent space at every $p \in S$:

$$\boxed{T_pS = \ker J(p)} \tag{24}$$

Proof sketch. By the implicit function theorem, near p the level set S is the graph of $f : V \to W$. A curve on the graph has the form $t \mapsto (x(t), f(x(t)))$, with velocity $(\dot{x}(0), Df(x_0)\dot{x}(0))$ — a vector space of dimension n. Since $\ker J(p)$ also has dimension n by rank-nullity, and $T_pS \subseteq \ker J(p)$, equal finite-dimensional spaces with one included in the other must coincide. The fully explicit construction is in [1]. ■

Remarks

- This theorem is our first necessary and sufficient condition to define a smooth submanifold.
- This theorem is local: the implicit function theorem gives a graph representation only in a neighborhood of p, and the regularity condition $\text{rank}\, J(p) = r$ must hold at each point separately.
- If instead $\text{rank}\, J(p) < r$ at some point p, the constraints are degenerate there: some constraints are redundant (linearly dependent at p), and the level set may develop a singularity (topologically, we can imagine a corner, a crossing, a cusp... any point where the manifold fails to be locally diffeomorphic to $\mathbb{R}^n$). Such points are called **singular points** and breaks the definition of a smooth submanifold.

Example: the sphere revisited. Once again, consider $F(x, y, z) = x^2 + y^2 + z^2$. From our new knowledge, we consider the associated Jacobian matrix, $J = (2x, 2y, 2z)$. This 1×3 matrix has rank 1 everywhere except at the origin (the null vector does not span any vector space). Since the origin is not on $F^{-1}(R^2)$ for any radius $R \neq 0$, every point of the sphere is regular. The tangent space at $p = (x_0, y_0, z_0)$ is $T_pS^2 = \ker(2x_0, 2y_0, 2z_0) = \{v \in \mathbb{R}^3 : x_0v^1 + y_0v^2 + z_0v^3 = 0\}$. This is exactly the plane tangent to the sphere at p: vectors perpendicular to the radius. The dimension is $3 - 1 = 2$.

Remark: on the choice of which coordinates are "free". The implicit function theorem guarantees that some $r \times r$ submatrix of $J(p)$ is invertible equivalently, that some r coordinates can be expressed in terms of the remaining n. Which ones depends on p and may vary across the manifold. This is why we speak of "free" and "dependent" coordinates only locally. Charts are really the object we use to define which are the free and the dependents coordinate for a submanifold.

D. Charts and atlases

Charts and atlases formalize what "locally looks like $\mathbb{R}^n$" means. Indeed, the implicit function theorem tells us that near any regular point p, the submanifold S looks like a copy of $\mathbb{R}^n$: we can parametrize it by n free coordinates $(x^1, \ldots, x^n)$. This is called a chart.

Definition: Chart

A chart on a submanifold S^n around a point p is a pair (U_p, φ) where:

- $U_p \subset S$ is an open neighborhood of p in S,
- $\varphi : U_p \to V_p \subset \mathbb{R}^n$ is a homeomorphism (a continuous bijection with continuous inverse) onto an open set V_p of $\mathbb{R}^n$,
- φ and φ^{-1} are smooth.

The map φ assigns n real numbers $(x^1(q), \ldots, x^n(q))$ to each point $q \in U$. These numbers are the local coordinates of q in the chart. The inverse $\varphi^{-1} : V \to U$ is called a local parametrization of S.

In the graph view, the local parametrization is exactly $\varphi^{-1}(x^1, \ldots, x^n) = (x^1, \ldots, x^n, f^1(x), \ldots, f^r(x))$. The free coordinates x^i are the chart coordinates.

Definition (Atlas)

A collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ such that $\bigcup_\alpha U_\alpha = S$ is called an atlas of S. Every submanifold admits an atlas: the implicit function theorem guarantees a chart around every regular point.

Definition: transition maps

When two charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) overlap (i.e. $U_\alpha \cap U_\beta \neq \emptyset$), we introduce the associated transition map as:

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta) \tag{25}$$

Compatibility condition

The compatibility condition assumes that all transition maps are smooth between open subsets of $\mathbb{R}^n$. In other words, the coordinates assigned by two overlapping charts must be smoothly related to each other.

Example: the circle S^1. Two charts suffice. Let $U_+ = S^1 \setminus \{(0, -1)\}$ (all but the south pole) and $U_- = S^1 \setminus \{(0, 1)\}$ (all but the north pole). On U_+, use the angle $\theta \in (-\pi, \pi)$; on U_-, use $\theta \in (0, 2\pi)$. These two charts cover S^1. On the overlap, the transition map is a smooth reparametrization of the angular coordinate. Alternatively and more concretely: the upper and lower semicircle graph representations we defined earlier are essentially two charts. The transition map on their overlap relates $x \mapsto +\sqrt{1-x^2}$ and $x \mapsto -\sqrt{1-x^2}$ near the equator: it is just $x \mapsto x$, which is smooth.

Connection to analytical mechanics

Generalized coordinates are charts. In Lagrangian mechanics, the configuration space of a mechanical system is a manifold Q (e.g. for a double pendulum, $Q = S^1 \times S^1$, a torus). Generalized coordinates $(q^1, \ldots, q^n)$ are precisely the coordinates of a chart $\varphi : U \subset Q \to \mathbb{R}^n$. The Euler-Lagrange equations are written in a particular chart; their form under change of generalized coordinates is the transition map. The fact that physics does not depend on the choice of chart is the coordinate-independence of the Euler-Lagrange equations, which you get for free from the variational formulation.

E. Vector fields

To conclude this section we will define an extremely important concept for all physics: a vector field.

Definition: tangent bundle

The collection of all tangent spaces of an n dimensional submanifold S is called the tangent bundle:

$$TS = \{(p, v) \mid p \in S,\ v \in T_pS\} \tag{26}$$

This is itself a manifold of dimension $2n$: near each point, a chart on S gives n coordinates for the base point p, and n coordinates for the vector v in the basis induced by that chart.

Definition: vector field.

A vector field on S is a smooth map:

$$\begin{aligned} X : S &\to TS, \\ p &\mapsto X_p \in T_pS \end{aligned} \tag{27}$$

That assigns to each point p a tangent vector at p. In a chart φ with coordinates $(q^1, \ldots, q^n)$, a vector field takes the form:

$$X = X^i(q)\frac{\partial}{\partial q^i} \tag{28}$$

Where the $X^i : V \subset \mathbb{R}^n \to \mathbb{R}$ are smooth functions. The basis vectors $\frac{\partial}{\partial q^i}$ are ;**the coordinate vector fields induced by the chart**: the velocity vectors of the coordinate curves $t \mapsto \varphi^{-1}(q^1, \ldots, q^i + t, \ldots, q^n)$.

In other words, a vector field is a map that "automatically" outputs tangent vector for any point of the submanifold. What is particularly powerful is that it *can* but *does not have to* be induced by charts. In physics, this means it is an object that **does not depend on the coordinate system**.

Smoothness.

X is smooth if and only if for every smooth function $f : S \to \mathbb{R}$, the function:

$$\begin{aligned} &Xf : S \to \mathbb{R} \\ &(Xf)(p) = X_p(f) = X^i(p)\frac{\partial f}{\partial q^i}(p) \end{aligned} \tag{29}$$

Is also smooth. This definition is coordinate-independent: it does not depend on which chart you use to compute it.

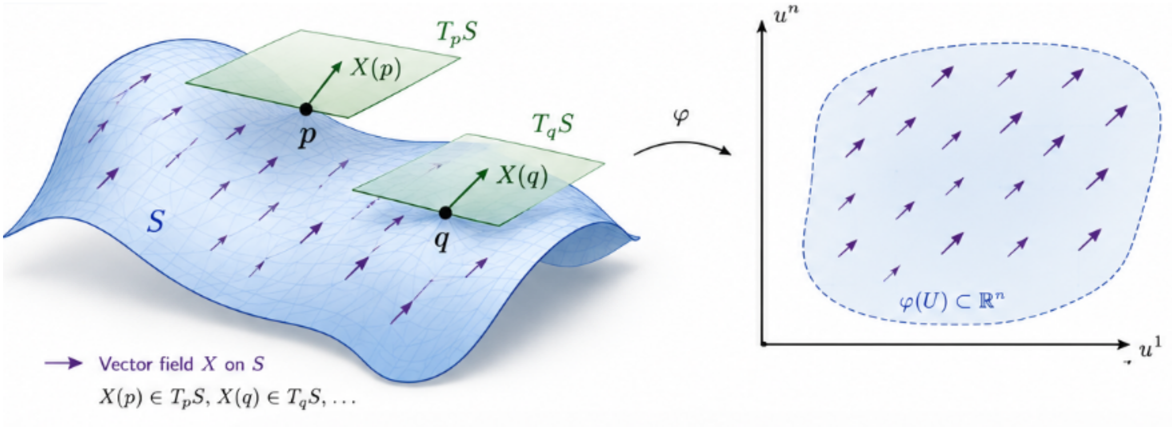

Figure 3: Illustration of a vector field on the submanifold: it is a map that output a vector from the tangent space near the point of evaluation in the submanifold (we showed examples at two points p and q). On the right we apply a chart to define locally the vector field. Picture generated by A.I.: feel free to suggest any picture that may represent better the concept of vector fields.

Connection to mechanics

In Newtonian mechanics, Newton 2nd law defines a vector field on the tangent bundle $T\mathcal{S}$ of the state space $\mathcal{S} = (r, \dot{r})$. As Newton equations are defined over the full state space, we can even express the corresponding associated vector field:

$$X_N : \mathcal{S} \to T\mathcal{S} s = (r, \dot{r}) \mapsto (\dot{r}, \frac{1}{m} \sum_k F_k) \tag{30}$$

A curve (called trajectory in this context) $\gamma : \mathbb{R} \to T\mathcal{S}$ such that

$$\partial_t \gamma = X_N \tag{31}$$

Is a solution of the Newtonian vector field: it solves the equation of motion induced by X_N and is therefore the physical trajectory taken by the system in the regime of validity of Newtonian mechanics.
Similarly, in Lagrangian mechanics, the Euler-Lagrange equations define a vector field on the tangent bundle TQ of the configuration space Q. A solution trajectory $q(t)$ is an integral curve of this vector field: a curve $\gamma : \mathbb{R} \to TQ$ such that $\dot{\gamma}(t) = X_{\gamma(t)}$ at every t.
The existence and uniqueness of integral curves (at least locally) follows from the Cauchy-Lipschitz theorem applied to the ODE $\dot{\gamma} = X \circ \gamma$. This is the precise sense in which a mechanical system defines a flow on its state space.

1.1.2 Abstract manifold

Everything above defined manifolds as subsets of an ambient $\mathbb{R}^{n+r}$. For most of classical and quantum mechanics, this is sufficient: configuration spaces, phase spaces, and constraint surfaces all arise naturally as submanifolds of some Euclidean space.

However, two situations force us to go further:

1. Intrinsic geometry: when one wants to define curvature, distances, or geodesics purely from within the manifold, without reference to how it sits in an ambient space. General relativity is the canonical example: spacetime is not naturally embedded in any higher-dimensional flat space.

2. Quotient constructions: some manifolds arise as equivalence classes (e.g. the space of rotations

$SO(3)$, or the projective space $\mathbb{RP}^n$) and have no natural ambient embedding.

An abstract manifold is defined by specifying an atlas directly: a collection of charts with smooth transition maps, without requiring any ambient space. Whitney's embedding theorem guarantees that any smooth abstract manifold of dimension n can be embedded in $\mathbb{R}^{2n}$, so the two approaches are equivalent in principle. But the intrinsic definition is more natural for many purposes.

Vector fields as derivations.

The map $X : \mathcal{C}^\infty(S) \to \mathcal{C}^\infty(S)$ is linear and satisfies the Leibniz rule:

$$X(fg) = f \cdot Xg + g \cdot Xf \tag{32}$$

This is the intrinsic, coordinate-free characterization of a vector field: it is a derivation on the algebra of smooth functions. On abstract manifolds, this is taken as the definition.

When we reach symplectic geometry and Hamiltonian mechanics, we will need the abstract definition to handle phase spaces and the space of rotations $SO(3)$ intrinsically.

This section will be completed later.

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